

Data Analytics for ICT Business

Combined Question Bank — Deduplicated

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Section 1

Part – A: Mandatory to Answer

- a. Find the median, lower quartile, upper quartile and inter-quartile range of the following data set of scores: 19, 21, 23, 20, 23, 27, 25, 24, 31. (CLO1) [2 Marks]
- b. A dataset “PQS” relates advertising expenditure (X) in thousands of BDT to sales (Y) in units. The regression equation is: [2 Marks]

$$Y = 50 + 8X$$

- Predict sales, when advertising expenditure is 5000 BDT.
 - Interpret the slope and intercept in the context of the given problem.
- c. In a given dataset “PQS.csv” that has both numerical and categorical attributes, how would you deal with missing values? Describe how various tactics affect the model’s performance. (CLO3) [2 Marks]
- d. How do outliers affect statistical analysis and machine learning models, and what techniques can be employed to determine whether to eliminate or keep them? (CLO2) [2 Marks]
- e. How can resource allocation and ICT service delivery be enhanced by predictive analytics? [2 Marks]
- f. A given dataset “QPR” contains 11 points with two classes (G and B). You are asked to classify a new point using KNN. Discuss the impact of choosing K such as K = 1 vs K = 7. (CLO3) [2 Marks]
- How does it affect the classification results?
 - Which K value is more robust against noise (error), and why?

Part – B: Answer any Four (from Question 2–7)

Question 2

- a. Define big data. Explain how big data can be converted into a business insight. [3 Marks]
- b. How is structured data different from unstructured data? [3 Marks]
- c. Discuss the relationship between business strategy and business analytics. [3 Marks]
- d. The following are the sales details of a company to X and Y for 2022. Plot a line graph using the data and comment on the sales trends. [3 Marks]

Particulars	Jan	Feb	March	April	May	Jun
X	140	135	140	155	170	189
Y	220	215	210	205	190	172

Question 3

- a. Explain how machine learning techniques are used for business data analytics. [4 Marks]
- b. How can you distinguish supervised and unsupervised learning in a machine learning model? [3 Marks]

- c. The following distribution gives the pattern of overtime work done by 100 employees of a company. Find the mean and median. [5 Marks]

Overtime (hrs.)	10–15	15–20	20–25	25–30	30–35
No. of Employees	11	20	35	20	18

Question 4

- a. Two systems are given as follows. Find out the availability for both systems and determine which system is more reliable in terms of availability.

System	MTTF (hrs)	MTTR (hrs)
P	900	100
Q	500	25

- b. A product costs 75 BDT to manufacture and has initial failure intensity $\lambda = 0.015$ failures/month. The warranty period is 18 months. There is a per-customer fixed support overhead of 1 BDT and 35 BDT per failure. Find out the unit cost with warranty. [3 Marks]
- c. A renowned software company plans to launch a new mobile banking application within 10 months, where the estimated budget is 250,000 BDT. Under these circumstances, create a Project Charter for the given project that includes the essential components of an IT project. [3 Marks]
- d. List out the reliability metrics. [2 Marks]

Question 5

- a. The following table shows retail transaction data (invoice, branch, city, customer type, product line, pricing, and rating fields). [7 Marks]
- Write a Python solution that fills the blank cells of “Unit price” using the mean and mode functions.
 - Write a Python solution that replaces the city name “Yangon” with “YG” in the City column.
 - Write a Python solution that fills the blank cell of “Tax 5%” using the calculation of Unit price and Quantity.
 - Visualize a scatter plot from the given table based on the columns City and Rating.
- b. An ICT company plans to launch a cloud-based education platform in emerging sustainable markets. Provide a SWOT analysis of the above business model. [3 Marks]
- c. Define activity-on-arrow network in terms of ICT business. [2 Marks]

Question 6

- a. Using the following dataset, find the class (KNN classification) of a new patient with (Age = 38, Blood Pressure = 135, Risk Level = ?) using $K = 5$. [3 Marks]

Age	Blood Pressure	Risk Level
25	120	Low
35	130	Low
45	150	High
50	160	High
30	125	Low
40	145	High

- b. Evaluate the following dataset and predict the fruit for a new sample with features (Weight = 160, Taste = 6) using K = 3, under K-Means clustering. [4 Marks]

Weight	Taste	Fruit
150	8	Coconut
130	7	Coconut
200	4	Pine-apple
180	5	Pine-apple

- c.i. How do you distinguish binary and multiclass classification? [3 Marks]
- c.ii. List out the data mining methods.
- d. Using an example, clarify the difference between prescriptive and predictive analysis. [2 Marks]

Question 7

PinkCity is a mid-sized metropolitan city aiming to reduce energy costs and carbon emissions while maintaining reliable power supply. Its energy department manages multiple sources — solar panels, wind turbines, and conventional power plants — but faces challenges: hourly and seasonal demand fluctuation, wasted energy from overproduction versus blackouts from underproduction, costly and limited battery storage, and high peak-hour electricity costs affecting residents and businesses.

- a.i. What kind of analytics system may be adopted to optimize energy production, storage, and distribution for PinkCity? [3 Marks]
- a.ii. Justify your answer to (i).
- b.i. Distinguish between correlation and regression analysis. [3 Marks]
- b.ii. List out applications of correlation and regression with examples.
- c.i. Define a data mart. Distinguish between a dependent data mart and an independent data mart with suitable examples. [3 Marks]
- c.ii. Write 2–3 examples of first-party data (collecting data) and justify. Also: in a survey, respondents were asked for their favorite fruit — Apple, Banana, or Mango. What type of categorical data is this, and why?
- d.i. KNN is considered a “lazy learner” or “eager learner.” What does this mean in practice, and what are the advantages and disadvantages of this property? [3 Marks]
- d.ii. How does the choice of visualization (scatter plot, histogram, box plot, heatmap) impact your understanding of a dataset? Give an example where the wrong choice could mislead interpretation.

Section 2

Part – A: Mandatory to Answer

- (a) i. How can one claim a machine learning model is good (efficient) or not good? [3 Marks]
- (a) ii. Consider the following data table:

Unit (x)	5	7	12	16	20	24
Price (y)	40	120	180	210	240	?

- Find out the regression line.
- Find the “?”.

(b) Assume a CSV file where the last six months' sales information is available. From sale.csv, use the column of total price per month. [2 Marks]

— Write Python code to import the data from the CSV file.

— Draw a histogram based on total price per month using Python.

(c) Consider five points $\{X_1, X_2, X_3, X_4, X_5\}$ as a two-dimensional sample for clustering: $X_1 = (0, 2.5)$; $X_2 = (0, 0)$; $X_3 = (1.5, 0)$; $X_4 = (5, 0)$; $X_5 = (5, 2)$. Compose the K-means partitioning algorithm using the above dataset. [3 Marks]

(d) Fill in the blanks of the ML workflow: Model Building $\rightarrow$ _____ $\rightarrow$ Model Testing $\rightarrow$ _____ $\rightarrow$ Applying Model $\rightarrow$ _____ [2 Marks]

(e) How do you handle missing or corrupted data in a dataset (using Python)? [2 Marks]

Part – B: Answer any Four Questions

Question 1

(a) i. Define data analytics and state the steps of the data analysis process. [3 Marks]

(a) ii. Deduce the applications of data cleaning.

(b) i. "Earth Data (by NASA)" — mention the type/category of the dataset, with justification. [3 Marks]

(b) ii. Explain the second-party data set using certain examples.

(c) Based on the dataset $\{1, 3, 6, 7, 12\}$, find the standard deviation and variance. Also find the range and coefficient of range of the dataset. [4 Marks]

(d) How do training and testing data work in machine learning? [2 Marks]

Question 2

(a) i. Using an example, describe the function of an outlier. [3 Marks]

(a) ii. Is it a good idea to follow a hierarchy of descriptive and predictive analytics before applying prescriptive analytics?

(b) i. How can analytics aid in objective decision making?

(b) ii. Define support vector machine (SVM), including the types of SVM.

(c) i. Short notes on OLAP and OLTP. [3 Marks]

(c) ii. Define big data with certain applications.

(d) How can you distinguish supervised and unsupervised learning in a machine learning model? [2 Marks]

Question 3

(a) Provide a detailed SWOT analysis for a joint-venture IT (software company) business.

(b) Given a discount rate of 10%, with costs of 1,00,000 BDT in year 0 and 20,000 BDT in years 1–3, and benefits of 0 BDT in year 0 and 1,60,000 BDT in years 1–3, find out the ROI. [4 Marks]

(c) Create a Three-Sphere Model for a project of purchasing smartphones for "PQS" college students.

(d) Write the tasks of data mining using certain examples.

Question 4

(a) i. Discuss data marts. [3 Marks]

(a) ii. Deduce the major characteristics and objectives of data mining.

(b) i. How is training data employed to teach ML and AI models? [3 Marks]

(b) ii. Give an example of image annotation.

(c) Are over-fitting and under-fitting good cases for a machine learning algorithm? Justify your answer.

[2 Marks]

(d) A linear regression model predicts car prices based on the model of the car. Ten data points are given with the following actual prices (x) and predicted prices (xp): [4 Marks]

x = [100, 150, 200, 250, 300, 350, 400, 450, 500, 550]

x_p = [110, 140, 180, 240, 290, 320, 380, 420, 480, 520]

- Find out the Root Mean Squared Error (RMSE).
- Find out the Mean Absolute Error (MAE).
- Find out R² (R-squared).

Question 5

(a) What is classification? What are the differences between classification and clustering? [3 Marks]

(b) Using certain examples, explain lazy learner and eager learner.

(c) Evaluate the following table and, using the KNN algorithm, find the class of (Brightness = 20, Saturation = 35, Class = ?), where K = 5. [4 Marks]

Brightness	Saturation	Class
40	20	Red
50	50	Blue
60	90	Blue
10	25	Red
70	70	Blue
60	10	Red
25	80	Blue

(d) Define linear regression and write the properties of linear regression. [2 Marks]

Question 6

(a) For example, a web-server installation was 3/4th completed by the end of week 1. The rate of performance would be 75%, since by the end of week 1 the planned schedule reflects that the task should be 100% complete, and only 75% of that work has been completed. If the planned value is 20,000 for week 1 and the actual cost is 25,000, find out the Earned Value, Cost Variance, Schedule Variance, CPI (Cost Performance Index), and SPI (Schedule Performance Index). [3 Marks]

(b) How does KNN work? Give an example and explain in your own words. [2 Marks]

(c) i. Define Hamming distance using an example. [4 Marks]

(c) ii. Seven observations are to be clustered into 3 clusters using K-Means clustering. After the first iteration, the clusters are:

- C₁: {(2,2), (4,4), (6,6)}
- C₂: {(0,4), (4,0)}
- C₃: {(5,5), (9,9)}
- What is the Manhattan distance for observation (9, 9) from the C₁ centroid in the second iteration?

(d) Explain how the regression process is used in data analysis. List the characteristics of linear regression.

[3 Marks]

Section 3 — Quiz 1

Marks: $3 \times 5 = 15$

Question 1

- a. Write down the steps of the data analysis process.
- b. i. Write 2–3 examples of first-party data (collecting data) and justify.
- b. ii. In a survey, respondents were asked for their favorite fruit: Apple, Banana, or Mango. What type of categorical data is this, and why?
- c. Find the variance and standard deviation of the following numbers: 1, 2, 4, 5, 7, 8, 9, and 11.

Question 2 (7 + 6 + 2)

- a. The sales of a company (in million dollars) for each year are shown in the table below. Use the least-squares regression line as a model to estimate the sales of the company in 2012. [7 Marks]

Year (x)	2005	2006	2007	2008	2009
Sales (y)	12	19	29	37	45

b. Study the dataset and solve these (use Python):

- i. Show the first 15 rows of the given dataset.
- ii. From the “Salary” column, find the wrong values where the value is greater than 1,00,000, then correct those values by replacing them with 55,000.
- iii. In the “Salaries” column, find the blank cells and fill them using the mean value.
- c. Define a data mart. Distinguish between a dependent data mart and an independent data mart with suitable examples. [2 Marks]

Question 3

- a. How does the choice of k affect the bias-variance tradeoff in KNN classification, and what risks arise when k is too small or too large? [3 Marks]
- b. KNN is considered a “lazy learner” or “eager learner.” What does this mean in practice, and what are the advantages and disadvantages of this property? [3 Marks]
- c. Given points {1, 3, 5, 7, 9, 11} with classes {B, A, B, A, B, A}. For test point $x = 7$, determine the predicted class for $k = 1$, $k = 3$, and $k = 5$. [5 Marks]
- d. i. Using an example, state the triple constraint of IT project management. [4 Marks]
- d. ii. List out the core functions of IT project management.

Section 4

Question 1

A large hospital uses a digital system to manage patient records, diagnoses, and billing. The data quality team discovered several anomalies while preparing reports: some patients have incorrect blood pressure readings (e.g., extremely high or low values that are medically unlikely); the same patient sometimes appears under slightly different IDs due to spelling variations in names; laboratory test results are sometimes uploaded late, causing reports to show outdated patient conditions; fields such as patient address and contact number are missing in many records; different departments use different formats for recording gender (e.g., “M”, “Male”, “1”); and some treatment dates are recorded after the discharge date, which is logically impossible. The hospital relies heavily on this data for treatment decisions, medical research, and billing operations.

- a. Identify possible types of data quality issues, with justifications. [5 Marks]
- b. Explain first-party data with an example. [5 Marks]
- c. Customer data is collected from multiple sources (CRM, website logs, surveys).
 - What challenges might arise in the Prepare step?
 - Suggest at least three methods to ensure the data is ready for analysis.

Question 2 (5 + 5)

A software company was hired by a university to develop an online learning management system within 6 months and a budget of BDT 80,000. The initial scope included student registration, online classes, attendance tracking, and exam management. After three months, the university requested additional features — a mobile application, AI-based student performance analysis, and online payment integration. At the same time, two experienced developers left the company, causing delays. To complete the extra features on time, the project manager considered hiring new developers and purchasing additional cloud services, which would increase overall cost; however, the university was unwilling to significantly increase the budget. The testing team also reported security issues requiring additional time to fix and verify. Facing pressure, the project manager had to decide whether to reduce scope, extend the deadline, or increase the budget. Finally, the university agreed to postpone the AI feature to a second phase, extend the deadline by two months, and slightly increase the budget — demonstrating the relationship among the three project constraints of scope, time, and cost, where changing one directly affects the other two.

- a. Based on the above case study, find out the constraints of project management with justifications.
- b. Explain how the Three-Sphere Business Model can be applied in designing a sustainable smart-city infrastructure project.
- c. What is a DataFrame in Pandas? Explain with an example.

Question 3 (7 + 3)

Answer the following based on the table below:

Student_ID	Name	Age	Marks	Department
101	Rahim	21	85	CSE
102	Karim	NaN	78	EEE
103	Anika	20	NaN	CSE
104	Suma	22	91	null
105	Rahim	21	85	CSE

- i. Load the dataset into Pandas.
- ii. Find the missing values.
- iii. Replace missing Age values with the average age.
- iv. Replace missing Marks values with the median marks.
- v. Replace null Department values with “Unknown”.
- vi. Remove duplicate rows.
- vii. Display the cleaned dataset.